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. These connected things need to talk to each other in order to learn from each other and correct themselves."

T. Anand, principal consultant, Knewron, points out that a lot of the software used in the electronics of things is indeed open source. "That means it can be easily modified to suit your specific usage needs. It gives (qualified) users enough power to improve the product and align it with users' day-to-day activities," he explains.

Communication protocols. The popular communication protocols for such embedded electronics are connection-oriented protocols, polling, time division multiple access, etc. TCP/IP is very popular for Ethernet-based systems. If the end nodes are connected via wireless, then ZigBee, Sub-GHz RF are also gaining. In some development environments, the increase may also be driven by the need to ease system integration burdens amongst multiple design groups or to provide system flexibility through smart sensors and smart actuators," says Tyagi.

Vishwanathan explains that there is a lot of debate on which communication protocols are ideal for everyday objects to communicate and become a part of the Internet. "There is no standard way for this to happen yet, and there are two levels at which the problem occurs. One is at the communication infrastructure level, which is whether they should use Wi-Fi, 4G, Bluetooth Smart, NFC or others. The second one, beyond the network stack to use for communication, is also the application-level protocol to use, such as http, XMPP, etc."

Speaking at the communication infrastructure level, Jayraj Ugarkar, head, Internet of Things Centre of Excellence at Infosys Labs lists, ZigBee, Bluetooth low energy (BLE), NFC, Flutter and LoRa. Flutter and LoRa are new ones that address the range limitations in ZigBee. BLE is very useful for a small range of applications and it gives a very long battery life. Each of these protocols is useful in different applications. For example, ZigBee Smart Energy profile is heavily used in appliances that participate in demand-response programs in smart grid as it provides PKI level of security. BLE is used when multiple devices need to communicate within a personal area network. Flutter

and LoRa are for long-range wireless communication as needed in smart city-based applications."

Vipin expresses his opinion that, "The L3 protocol of choice is IP. Protocols at L4 and above tend to be application driven. A protocol that, in my opinion, towers amongst all is https. Https is ubiquitous and is considered to be secure enough for most embedded applications."

In the long run, it is likely to turn out as Natarajan says, "Wireless communication protocols covering short, medium and long distances combined with vertical domain protocols, eg, building management, energy, healthcare, etc are expected to see more and more adoption, standardisation and improvements."

Tools galore. Electronics of things, as we can see is characterised by hardware, software and networks that meet special requirements. As a result, it also requires specialised platforms and tools for easy development and deployment—a space that industry players are beginning to tap with vigour.

For example MathWorks Inc. has been providing MATLAB and Simulink along with specific toolboxes, block sets and coders for algorithm development, system design, prototyping and implementation and verification of embedded software and hardware that have been widely adopted across many industries and application areas. National Instruments provides advanced tools for testing such systems. "Hardware-in-loop testing is becoming more and more prevalent to meet the highest quality of standards for these products. Hardware and software tools being used are again platform based. One such platform that is widely accepted is the graphical system design platform based on the LabVIEW RIO architecture, which enables such sophisticated testing that it involves deterministic execution of models in real-time to recreate real-life scenarios for testing these electronic equipment," says Satish Mohanram, technical marketing manager, National Instruments.

Others like Infosys and Persistent Systems work across the value chain, whereas few others like Knewron provide guidance and consultancy for developing electronics/IoT products and platforms. Indeed, a happening space with individual users, innovators, companies and solution providers is active! We can look forward to further interesting developments in this field.

now almost everything has at least one microcontroller inside it and programmability has increased manifold; which means one can easily hack into certain electronic gadgets or equipment and experiment with them. Moreover, all the electronic parts are easily available nowadays at relatively affordable prices. This phenomenon (of programmability and availability) has fuelled creative thinking in terms of usage and applications of electronics in daily life. Makers now can experiment with (less expensive) electronics and realise their imagination."

Increasing comfort levels. The proliferation of smartphone could also be the reason for people's increasing comfort (and dependency) in using technology for their everyday tasks. "Thanks in part to the smartphone revolution with its emphasis on apps, there has been a fundamental shift in the way people see electronic interfaces. They are also able

to see the value that electronics adds to their life—they can do things better, faster, with less effort, etc, since electronics can handle the bulk of the task that people had to do previously. For this improved ability, convenience and comfort, they are willing to pay a good price. On the supply side, manufacturers have been quick to realise that there is a growing market for electronics in equipment and have started mass-producing such products, improving innovation and driving down costs, further fuelling demand. This kind of virtuous cycle has helped the proliferation of electronics in everyday products and enabled an ecosystem to develop," explains Kashinath.

Framework provided by the IoT.

Whilst at present, the IoT is just a subset of the Electronics of Things, some experts, including Ugarkar feel that the IoT is fuelling the tendency to stitch electronics into everything.

Ugarkar feels that "The use of sensors, actuators, meters that you see getting embedded into objects around us is not new. The boom you see in embedding electronics to sense, act or meter is because the IoT gave a high-level conceptual architecture that made it easy for embedded electronics components to integrate with each other and with the rest of the world using Internet technologies like IPV6, http, REST, Cloud, network protocols, etc."

On the other hand, experts like Anand feel that 'requirement fuels innovation' and not the other way round. According to him, it is the proliferation of the electronics of things that is leading to the IoT.

Is not this simply embedded electronics?

What is so new about the Electronics of Things? Has not electron-